

B.C.A. SEM - VI

Subject: Web Application Development Using ASP.Net

UNIT 3

Introduction

- ◆ ADO.NET Architecture.
- ◆ Introduction of Server Explorer and its Features.
- ◆ Sqlexpress database Connectivity using ADO.NET (INSERT, UPDATE, DELETE, SELECT).
- ◆ Data control – Grid view & Data list.

Q.-1 Explain ADO.Net Features and Architecture in detail.**Introduction:**

- ❑ ADO.NET is a set of computer software components that programmers can use to access data and data services based on disconnected DataSets and XML.
- ❑ It is a part of the base class library that is included with the Microsoft .NET Framework.
- ❑ It is commonly used by programmers to access and modify data stored in relational database systems, though it can also access data in non-relational sources.
- ❑ ADO.NET is sometimes considered an evolution of ActiveX Data Objects (ADO) technology, but was changed so extensively that it can be considered an entirely new product.

Difference between ADO and ADO.NET

ADO	ADO.NET
Use the RecordSet object	Use the DataSet object
Involves one table per RecordSet. If we want to use more than one table we need to do inner join and fill the RecordSet	Allows multiple tables per DataSet.
Stores the data in the XML format.	Store the data in binary format.
Use OLEDB providers and ODBC drivers.	Uses the .NET data provider.
Use the old read-only and forward-only featured to read the data from the RecordSet.	Uses the DataReader object to read the data from data source.
Executes SQL statements directly.	Executes the SQL statements through a command object, but not directly.
Supports client-side and server-side cursors to handle the data of RecordSet.	Does not support cursors, as they are not required in disconnected data.
Supports database locking.	Does not support database locking.
It is based on COM (Component Object Modelling) .	It is a CLR (Common Language Runtime) based library .
It works only when data store is Connected .	It does not needs active connection to access data from data store.
It has feature of locking .	It does not have feature of locking .
It access and store data from data source by recordset object .	It access and store data from data source by dataset object .
XML integration is not feasible in ADO.	XML integration is feasible in ADO.NET.
In ADO, data is stored in binary form .	While in this, data is stored in XML .
It allows us to create client-side cursors only .	It gives us the choice of using client side and server-side cursors .
It requires SQL JOINS and UNIONS to combine data from multiple tables in a single result table.	It uses DataRelational objects, for combining data from multiple tables without requiring JOINS and UNIONS.
It supports sequential access of rows in a RecordSet .	It allows completely non- sequential data access in DataSet through collection-based hierarchy.

Features of ADO.NET:**❏ Disconnected data architecture:**

- ❏ ADO.NET uses the disconnected data architecture.
- ❏ Application connects to the database only while retrieving or updating data.
- ❏ After the data is retrieved, the connection with the database is closed; when the database needs to be updated, the connection is reestablished.

❏ Data cached in database :

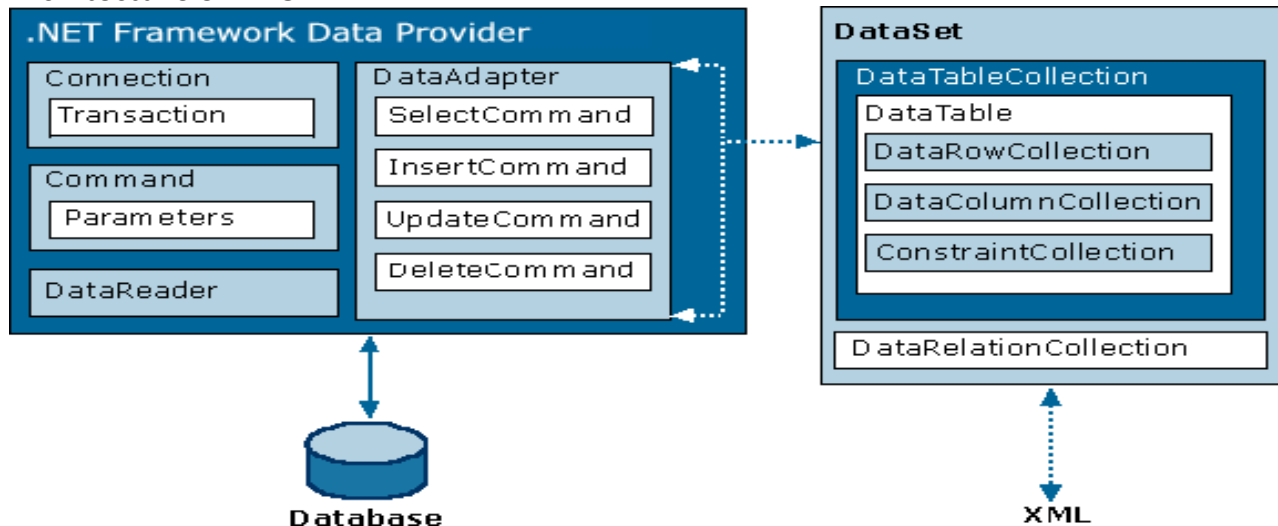
- ❏ A database is the most common method of accessing data since it implements a disconnected architecture.
- ❏ Since ADO.NET is based on a disconnected data structure, it is not possible for application to interact with the database for processing each record.
- ❏ Therefore, the data is retrieved and stored in the database.

❏ Data transfer in XML format:

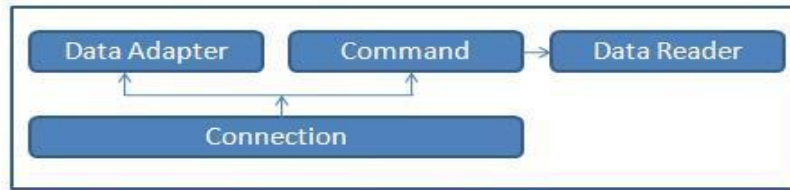
- ❏ XML is the fundamental format for data transfer in ADO.NET.
- ❏ Data is transferred from a database into a dataset and from another component by using XML.
- ❏ You can even use XML files as a data source and store data from it in a dataset.
- ❏ The knowledge of XML is hidden from the user.

❏ Interaction with the database is done through data commands:

- ❏ All operations on the database are performed by using data commands.
- ❏ Data command can be a SQL statement or a stored procedure.
- ❏ You can retrieve, insert, delete or modify data from a database by executing data commands.

Architecture of ADO.NET:**Components of ADO.NET:****❏ Data Providers**

- ❏ The two key components of ADO.NET are Data Providers and DataSet.
- ❏ The Data Provider classes are meant to work with different kinds of data sources.
- ❏ They are used to perform all data-management operations on specific databases.
- ❏ The .Net Framework includes mainly three Data Providers for ADO.NET.
- ❏ They are the Microsoft SQL Server Data Provider, OLEDB Data Provider and ODBC Data Provider.
- ❏ SQL Server uses the **SqlConnection** object, OLEDB uses the **OleDbConnection** object and ODBC uses the **OdbcConnection** object respectively.



- ❏ A data provider contains Connection, Command, DataAdapter, and DataReader objects.
- ❏ These four objects provides the functionality of Data Providers in the ADO.NET.

❏ Connection

- ❏ The Connection Object provides physical connection to the Data Source.
- ❏ The Connection Object is a part of ADO.NET Data Provider and it is a unique session with the Data Source.
- ❏ The Connection Object provides physical connection to the Data Source.
- ❏ Connection object needs the necessary information to recognize the data source and to log on to it properly, this information is provided through a connection string.
- ❏ The Connection Object connect to the specified Data Source and open a connection between the application and the Data Source, depends on the parameter specified in the Connection String.
- ❏ When the connection is established, SQL Commands will execute with the help of the Command Object and retrieve or manipulate data in the Data Source.
- ❏ Once the Database activity is over, Connection should be closed and release the Data Source resources.
- ❏ Two of the most common connection objects used is OleDbConnection and SqlConnection.
- ❏ The table given below display the commonly used properties and methods of a Connection object.

Name	Description
ConnectionString()	It provides information such as the data source and database name that is used to establish connection with database. ConnectionString Specifies Provider, DataSource, Integrated Security E.g. “Provider=Microsoft.Jet.OLEDB.4.0; DataSource=d:\abc.mdb”
Open() Method	It is used to open connection with the data source that is specified in the ConnectionString property.
Close() Method	It is used to close connection with data source.
State	It is used to describe the current state if the Connection object. The value 0 indicates that the connection is closed and value 1 indicates that the connection is open.
Dispose	It release the connection object

❏ Command

- ❏ The Command Object uses to perform SQL statement or stored procedure to be executed at the Data Source. The command object provides a number of Execute methods that can be used to perform the SQL queries in a variety of fashions.
- ❏ The Command Object requires an instance of an ASP.NET Connection Object for executing the SQL statements.
- ❏ The Command Object in ADO.NET executes SQL statements and Stored Procedures against the data source specified in the ASP.NET Connection Object.
- ❏ The Command Object has a property called CommandText , which contains a String value that represents the command that will be executed against the Data Source.
- ❏ When the CommandType property is set to StoredProcedure, the CommandText property should be set to the name of the stored procedure.

❏ Data Reader

- ❏ DataReader Object in ADO.NET is a stream-based , forward-only, read-only retrieval of query results from the Data Sources , which do not update the data.
- ❏ The DataReader cannot be created directly from code, they can created only by calling the ExecuteReader method of a Command Object.
- ❏ After creating an instance of the Command object, you have to create a DataReader by calling Command.ExecuteReader to retrieve rows from a data source.
- ❏ When the ExecuteReader method in the SqlCommand Object execute , it will instantiate a SqlClient.SqlDataReader Object.
- ❏ When we started to read from a DataReader it should always be open and positioned prior to the first record.
- ❏ The Read() method in the DataReader is used to read the rows from DataReader and it always moves forward to a new valid row, if any row exist .
- ❏ You should always call the Close method when you have finished using the DataReader object.
- ❏ Using data reader results in faster access to data and less memory usage since at any time, only a single row is stored in the memory. The table given below displays the common methods.

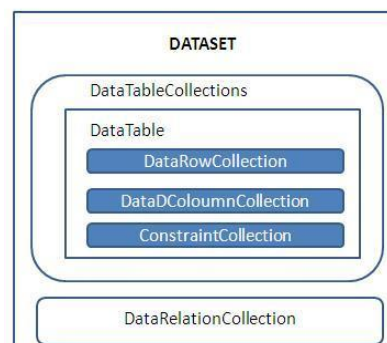
❏

NAME	DESCRIPTION
Read Method	It is used to read a row of data returned by executing the command on a data source.
Close Method	It is used to close the DataReader object.
NextResult Method	It is used to navigate to the next of data in case the command executing returns multiple rows of data.

❏ Data Adapter

- ❏ DataAdapter serves as a bridge between a DataSet and SQL Server for retrieving and saving data.
- ❏ We can use SqlDataAdapter Object in combination with Dataset Object.
- ❏ DataAdapter provides this combination by mapping Fill method, which changes the data in the DataSet to match the data in the data source, and Update, which changes the data in the data source to match the data in the DataSet.
- ❏ A data adapter retrieves data from a database into a dataset and updates the database.
- ❏ When you make changes to the dataset, the changes in the database are actually done by the data adapter.

- ❏ There are two types of data adapters that can be configured to connect a database on Visual Studio .NET.
 - **SqlDataAdapter:** This type of data adapter is configured to access data specifically from Microsoft SQL Server.
 - **OleDbDataAdapter:** This type of data adapter is configured to access data from any database that is supported by an OLEDB data provider.
- ❏ A data adapter handles data transfer between the database and the dataset through its properties and methods.
- ❏ Data Adapter Properties and Methods:
- ❏ A data adapter communication with a database while retrieving, inserting, deleting and updating data. The following properties and methods of data adapter can be set to perform various operations on a database:
 - **SelectCommand** – Refers to a SQL statement or a stored procedure to retrieve data from the database.
 - **InsertCommand** - Refers to a data command to insert data into a database.
 - **UpdateCommand** - Refers to a data command to update a database.
 - **DeleteCommand** - Refers to a data command to delete data from a database.
 - **Fill() Method** – Fills the dataset with the records from a database.
 - **Update() Method** – Executes the corresponding, InsertCommand, UpdateCommand or DeleteCommand for each insert, modified or deleted row to reflects the changes in the database.
- ❏ **Data Set**
 - ❏ DataSet provides a disconnected representation of result sets from the Data Source, and it is completely independent from the Data Source.
 - ❏ DataSet provides much greater flexibility when dealing with related Result Sets.
 - ❏ DataSet contains rows, columns, primary keys, constraints, and relations with other DataTable objects.
 - ❏ It consists of a collection of DataTable objects that you can relate to each other with DataRelation objects.
 - ❏ The DataAdapter Object provides a bridge between the DataSet and the Data Source.



Accessing data with Server Explorer, Binding Data**Q. 2 Explain Server Explorer and its Features in detail.****Introduction**

- ❑ Server explorer allows you to view the services that are available on a particulate server.
- ❑ Server Explorer is a powerful tool that provides the drag and drop features.
- ❑ It helps you to work databases in an easy way to use graphical environment.
- ❑ You can drag a service from server explorer onto a design surface, so that you can interact with the service programmatically inn your applications.
- ❑ Server Explorer is the server management console for visual studio.
- ❑ You can view and retrieve information from all the databases, you are connected to.
- ❑ You can use server explorer to list database tables, view, stored procedures and functions.
- ❑ Server Explorer is a powerful window in Microsoft Visual Studio that allows ASP.NET developers to view, manage, and interact with server-based resources directly from the IDE.
- ❑ It acts as a centralized interface for accessing databases, data connections, web services, servers, message queues, and other system resources used in ASP.NET applications—without writing code.
- ❑ In ASP.NET development, Server Explorer simplifies backend configuration and speeds up application development by providing a graphical way to connect and manage data sources and services.

Features of Server Explorer in ASP.NET**1. Database Connectivity**

- ❑ Connect to SQL Server, SQL Server Express, LocalDB, Oracle, and other databases
- ❑ View tables, views, stored procedures, and functions
- ❑ Create, modify, and delete database objects
- ❑ Preview and edit table data

2. Data Connection Management

- ❑ Add and manage multiple data connections
- ❑ Configure connection strings used in ASP.NET applications
- ❑ Test database connections directly

3. Drag-and-Drop Support

- ❑ Drag database tables or stored procedures onto ASP.NET pages
- ❑ Automatically generate data-bound controls and datasets

4. Web Services Access

- ❑ View and manage connected web services
- ❑ Explore service methods and properties

5. Server and System Resources

- ❑ Access event logs
- ❑ Manage message queues
- ❑ View performance counters
- ❑ Explore Windows services (depending on permissions)

6. Integration with Toolbox and Designers

- ❑ Seamless integration with Data Sources and Toolbox
- ❑ Helps in rapid application development (RAD)

7. Secure and Centralized Management

- ❑ Uses Windows authentication and secure credentials
- ❑ Centralized access to server-side components

Other Features of SQL Server

- ❑ Provide reliable storage.
- ❑ Provide a means to rapidly access the data.
- ❑ Ensure consistent access to the data
- ❑ Implement security
- ❑ Enforce data integrity; ensure the data stays consistent.

Server can perform following task:

- ❑ Open the data connection and commands.
- ❑ Logs on the servers and displays their databases and system services
- ❑ Views information about available web services and the methods
- ❑ Make data connection to sql servers.
- ❑ Stores database projects and references.
- ❑ Interacts with data resources.

Two primary nodes of Server Explorer:**❑ Data Connection :**

- ❑ List any data connections added by user to the server and is categorised as the top level node. You can make data connection to many types of databases.
- ❑ The various child nodes available in the Data Connections node are as follows :
 - ❑ Database diagrams
 - ❑ Tables
 - ❑ Views
 - ❑ Store procedures
 - ❑ Functions

❑ Server :

- ❑ List the currently connected servers and is available for use. It is categorized as the top level node. You can use the database and server resources in your application that are user each server node.
- ❑ The various child node types available in server nodes are as follow:
 - ❑ **Even log:** List application, security and system event logs on the selected server and Windows NT and any custom logs presents on the server.
 - ❑ **Management Classes:** Enable the developers to browse and modify Windows Management Instrumentation (WMI) data and call methods.
 - ❑ **Management Events:** Enable developers to register for WMI.
 - ❑ **Message Queues:** List the message queues, along with the messages that each queue contains on the selected server.
 - ❑ **Performance Counter:** List the performance counters on the selected server applications and websites.
 - ❑ **Services:** List the services that exists on the selected server. It also list the current status of each service.

Steps to connect with database

- ❑ Step-1 View of solution Explorer
- ❑ Step-2 Right Click on solutions explorer -> Add New Item
- ❑ Step-3 Add SqlServer database and rename it.
- ❑ Step-4 Database automatically stores in APP_DATA folder.
- ❑ Step-5 Server Explorer display our database and its' components
- ❑ Step-6 Right click on "TABLE" folder and right click on Add New Table
- ❑ Step-7 Set primary key and give name to the table.
- ❑ Step-8 Enter records in table.

Data Binding:

- ❑ Data binding refers to the process of binding web forms controls to data sources.
- ❑ It is done to display and access the data stored in a data source directly by an application.
- ❑ In other words, the control bound to a data source displays the information and data stored in that data source.
- ❑ For example, to display the names of the employees of a company in a ListBox control, you need to bind this control to the table containing the required data.
- ❑ Data that if you modify in a control, that change is also reflected in the original data source.
- ❑ In other words, data binding is a two way process in which the changes made in the data source are reflected in the control and vice versa.

Tasks performed by data binding are:

- ❑ **Navigating data** : refers to the process of moving from one record to another, based upon some specific search criteria.
- ❑ **Entering data** : refers to the process of entering data in a data entry form. The entered data is then sent to the database with which the data entry form is connected.
- ❑ **Creating master/detail relation** : refers to the process of binding the data of multiple tables together in a master and details relation. The master and details relation is a relation of tables where a parent table is associated with its child tables.
- ❑ **Performing data lookup** : refers to the process of looking up the data of a field corresponding to another field of the table. This type of data binding is supported by the ListBox control.

Two ways of data binding:

- ❑ Manual data binding:
- ❑ In this way, for binding controls of web form we have to set various properties, such as data source, data member, display member and value member. We will see example in detail in topic No. 4 in this unit.
- ❑ Visual Studio IDE data binding:
- ❑ Simple data binding:
- ❑ In this way, one in which you bind the control to a single data element.
- ❑ Complex data binding:
- ❑ Here you can bind the control to more than one record in the database. We will see example in detail in topic No. 5 in this unit with different data controls.

Database programming using code:**Database Programming using Code:**

- ❑ For database programming using code, we need to first add
Imports System.Data.SqlClient
Partial Class _Default
Inherits System.Web.UI.Page
- ❑ Imports name space called System.Data and then for SQL server, add SqlConnection for access, adds OleDb before any Class declaration.
- ❑ We will use System.Data.SqlClient and the System.Data namespaces of ADO.Net. The System.Data contains basic enumerations and classes, which we will use below.
- ❑ The System.Data.SqlClient provides data access to SQL servers such as MS SQL server 2000 and higher.

Connection Object

- ❑ The connection object creates connection to the database.
- ❑ The connection object represents the physical connection to a data source.
- ❑ The connection object contains all of the information required to open a connection to the database.

Connection object have connection string property

- ❑ A connection string has a group of semi-colon-separated attributes.
- ❑ Every .Net Data Provider connection string looks different; depending on the type of .NET data provider you need to use and which attributes are set each different type of database system.
- ❑ Depends on the parameter specified in the connection string, ADO.NET connection object connect to the specified database.
- ❑ The following examples show how to connect to a SQL Server database by using the SQL Client .NET data provider.

```
cn = New SqlConnection("Data Source=.\SQLEXPRESS;AttachDbFilename=D:\asp_net
examples\Connectivity\App_Data\Database.mdf;Integrated Security=True;User
Instance=True")
```

KEYWORDS	DESCRIPTION
Initial Catalog	Name of the database
Data Source	Server name or the connection of SQL server
Integrated Security	whether the connection is a secure connection.

Method of connection object

- ❑ **Open:** It opens the connection to our database
- ❑ **Close:** It closes the database connection
- ❑ **Dispose:** It release the resources on the connection object
- ❑ **State:** Returns 0 for not connected or 1 for connected. Return state of connection object.

Initializing connection string

```
Imports System.Data.Sql
Imports System.Data.SqlClient
ProtectedSub Page_Load(ByVal sender AsObject, ByVal e As System.EventArgs)
HandlesMe.Load
    cn = New SqlConnection("Data
Source=.\SQLEXPRESS;AttachDbFilename=D:\asp_net
examples\Connectivity\App_Data\Database.mdf;Integrated Security=True;User
Instance=True")
```

- ❑ We can perform three operations with database that are insert, update and delete.
- ❑ These three operation can perform by three ways (1) using sqldataadapter (2) using dataset (3) using sqlcommand
- ❑ Following example for database programming using code is perform above three operations using sqldataadapter:
- ❑ Below figure shows the output screen. Here we make a database "employee.mdf", we make "emp" table in the database which have empid, name, address and phone.

- ❑ In the design view we define 4 labels, 4 textbox and 3 buttons.

- Textboxes have name txtid for employee id, txtnam for employee name, txtadd for address and txtph for phone.

Visual basic code is given below (INSERT, UPDATE, DELETE, SELECT).

```
Imports System.Data.Sql
Imports System.Data.SqlClient
PartialClass _Default
Inherits System.Web.UI.Page
Dim cn AsNew SqlConnection
Dim da AsNew SqlDataAdapter
Dim ds AsNew DataSet
Dim query AsString
Dim a AsInteger
Dim b AsInteger
ProtectedSub Page_Load(ByVal sender AsObject, ByVal e As
System.EventArgs) HandlesMe.Load
    cn = New SqlConnection("Data
Source=.\SQLEXPRESS;AttachDbFilename=D:\asp_net
examples\Conectivity\App_Data\Database.mdf;Integrated Security=True;User
Instance=True")
    cn.Open()
EndSub

ProtectedSub Button1_Click(ByVal sender AsObject, ByVal e As
System.EventArgs) Handles Button1.Click
    query = "insert into emp values(" + Txtid.Text + "," +
Txtnam.Text + "," + TxtAdd.Text + "," + Txtph.Text + ")"
    da = New SqlDataAdapter(query, cn)
    da.Fill(ds)
EndSub

ProtectedSub Button2_Click(ByVal sender AsObject, ByVal e As
System.EventArgs) Handles Button2.Click
If Button2.Text = "Update"Then
Call dis()
    Button2.Text = "Save"
Else
    query = "update emp set empid=" + Txtid.Text + ", name=" +
Txtnam.Text + ", address=" + TxtAdd.Text + ", phone=" + Txtph.Text +
""
    da = New SqlDataAdapter(query, cn)
    da.Fill(ds)
    Button2.Text = "Update"
    MsgBox("Record Update")
Call clea()
EndIf
EndSub
PublicSub clea()
    TxtAdd.Text = ""
    Txtid.Text = ""
    Txtnam.Text = ""
    Txtph.Text = ""
EndSub
PublicSub dis()
    a = InputBox("Enter a id no to update a record")
    query = "select * from emp where empid=" + Str(a)
    da = New SqlDataAdapter(query, cn)
    da.Fill(ds)
    Txtid.Text = ds.Tables(0).Rows(0).Item(0)
    Txtnam.Text = ds.Tables(0).Rows(0).Item(1)
    TxtAdd.Text = ds.Tables(0).Rows(0).Item(2)
```

```
        Txtph.Text = ds.Tables(0).Rows(0).Item(3)
ExitSub
EndSub

ProtectedSub Button3_Click(ByVal sender As Object, ByVal e As
System.EventArgs) Handles Button3.Click
Call dis()
        b = MsgBox("Do you want to delete it", MsgBoxStyle.YesNo)
If b = 6 Then
        query = "delete from emp where empid=" + Str(a)
        da = New SqlDataAdapter(query, cn)
        da.Fill(ds)
Else
ExitSub
EndIf
EndSub
EndClass
```

Q.3 Explain Sqlexpress database Connectivity using ADO.NET.**Introduction to MS SQL Server:**

- ❑ Microsoft SQL Server is a relational database management system developed by Microsoft Inc.
- ❑ As a database, it is a software product whose primary function is to store and retrieve data as requested by other software applications.
- ❑ There are at least a dozen different editions of Microsoft SQL Server aimed at different audiences and for different workloads.
- ❑ Its primary query languages are T-SQL and ANSI SQL.

History:

- ❑ Prior to version 7.0 the code base for MS SQL Server was sold by Sybase SQL Server to Microsoft, and was Microsoft's entry to the enterprise-level database market, competing against Oracle, IBM, and, later, Sybase.
- ❑ Microsoft, Sybase and Ashton-Tate originally worked together to create and market the first version named SQL Server 1.0 for OS/2 (about 1989) which was essentially the same as Sybase SQL Server 3.0 on Unix, VMS, etc.
- ❑ Microsoft SQL Server 4.2 was shipped around 1992 (available bundled with IBM OS/2 version 1.3).
- ❑ Later Microsoft SQL Server 4.21 for Windows NT was released at the same time as Windows NT 3.1.
- ❑ Microsoft SQL Server v6.0 was the first version designed for NT, and did not include any direction from Sybase.

SQL Server Release History

Version	Year	Release Name	Codename
1.0(OS/2)	1989	SQL Server 1.0(16 bit)	-
1.1(OS/2)	1991	SQL Server 1.1(16 bit)	-
4.21(WinNT)	1993	SQL Server 4.21	SQLNT
6.0	1995	SQL Server 6.0	SQL95
6.5	1996	SQL Server 6.5	Hydra
7.0	1998	SQL Server 7.0	Sphinx
-	1999	SQL Server 7.0OLAP Tools	Palato mania
8.0	2000	SQL Server 2000	Shiloh
8.0	2003	SQL Server 200064-bit Edition	Liberty
9.0	2005	SQL Server 2005	Yukon
10.0	2008	SQL Server 2008	Katmai
10.25	2010	SQL Azure DB	CloudDatabase
10.5	2010	SQL Server 2008 R2	Kilimanjaro (aka KJ)
11.0	2012	SQL Server 2012	Denali
12.0	2014	SQL Server 2014	

Editions:**❏ Datacenter**

- ❏ SQL Server 2008 R2 Datacenter is the full-featured edition of SQL Server and is designed for datacenters that need the high levels of application support and scalability.
- ❏ It supports 256 logical processors and virtually unlimited memory.
- ❏ Comes with StreamInsight Premium edition.
- ❏ The Datacenter edition has been retired in SQL Server 2012, all its features are available in SQL Server 2012 Enterprise Edition.

❏ Enterprise

- ❏ SQL Server Enterprise Edition includes both the core database engine and add-on services, with a range of tools for creating and managing a SQL Server cluster.
- ❏ It can manage databases as large as 524 pet bytes and address 2 terabytes of memory and supports 8 physical processors. SQL 2012 Enterprise Edition supports 160 Physical Processors.

❏ Standard

- ❏ SQL Server Standard edition includes the core database engine, along with the stand-alone services.
- ❏ It differs from Enterprise edition in that it supports fewer active instances and does not include some high-availability functions such as hot-add memory, and parallel indexes.

❏ Web

- ❏ SQL Server Web Edition is a low-TCO option for Web hosting.
- ❏ Business Intelligence
- ❏ Introduced in SQL Server 2012 and focusing on Self Service and Corporate Business Intelligence. It includes the Standard Edition capabilities and Business Intelligence tools: PowerPivot, Power View, and the BI Semantic Model, Master Data Services, Data Quality Services and xVelocity in-memory analytics.

❏ Workgroup

- ❏ SQL Server Workgroup Edition includes the core database functionality but does not include the additional services. Note that this edition has been retired in SQL Server 2012.

❏ Express

- ❏ SQL Server Express Edition is a scaled down, free edition of SQL Server, which includes the core database engine.
- ❏ While there are no limitations on the number of databases or users supported, it is limited to using one processor, 1 GB memory and 4 GB database files.
- ❏ It is intended as a replacement for MSDE.
- ❏ Two additional editions provide a superset of features not in the original Express Edition.
- ❏ The first is SQL Server Express with Tools, which includes SQL Server Management Studio Basic.
- ❏ SQL Server Express with Advanced Services adds full-text search capability and reporting services.

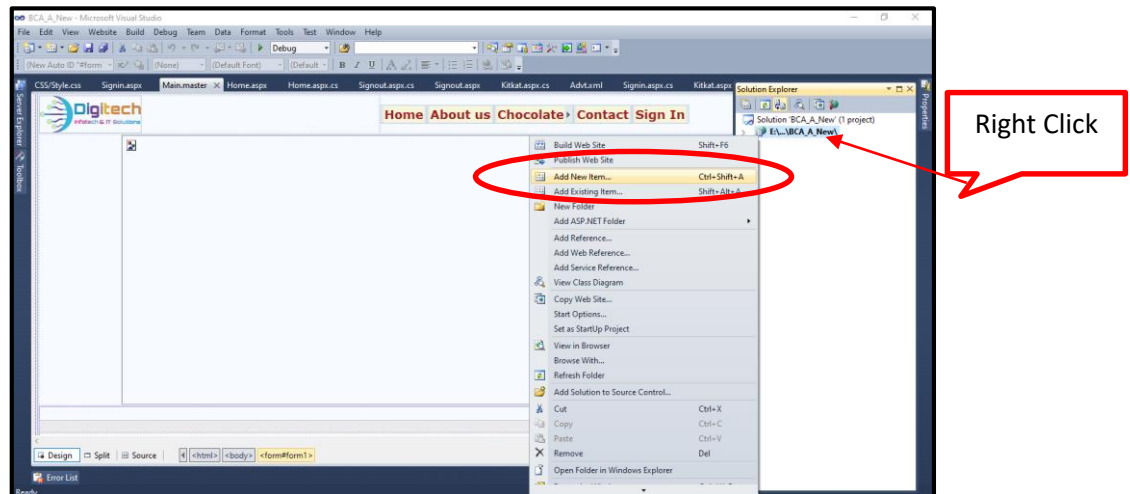
Architecture

- ❏ The protocol layer implements the external interface to SQL Server.
- ❏ All operations that can be invoked on SQL Server are communicated to it via a Microsoft-defined format, called Tabular Data Stream (TDS).
- ❏ TDS is an application layer protocol, used to transfer data between a database server and a client.

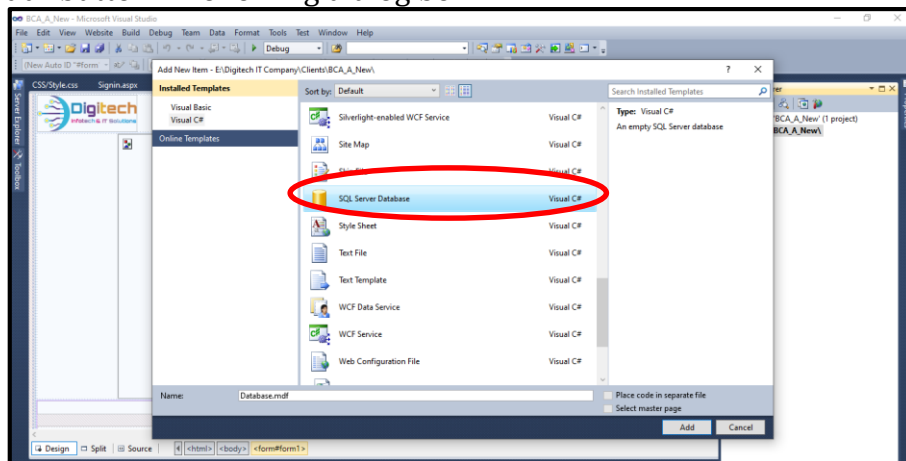
- Initially designed and developed by Sybase Inc. for their Sybase SQL Server relational database engine in 1984.
- Later by Microsoft in Microsoft SQL Server, TDS packets can be encased in other physical transport dependent protocols, including TCP/IP, Named pipes, and Shared memory. Consequently, access to SQL Server is available over these protocols. In addition, the SQL Server API is also exposed over web services.

Steps to create a database in MS SQL server:

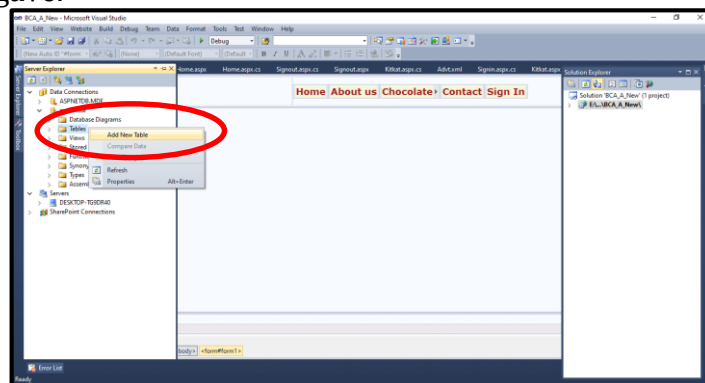
- Open a website.
- Now you have to right click on the path of web site in the “solution explorer”.
- Now you have to select “Add New Item” from the above menu.



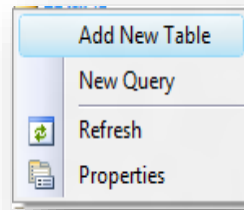
- Now you have to select “SQL Server Database” define appropriate database name and click on “Add” button in following dialog box.



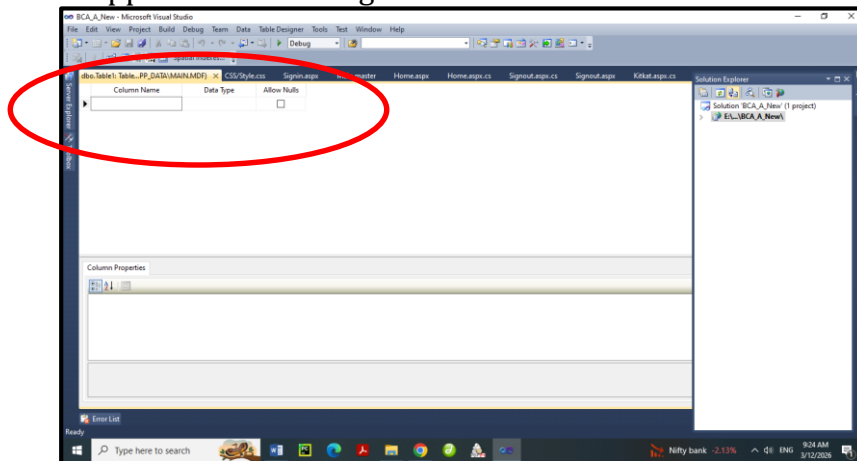
- So, “server Explorer” window will open show as bellow. You can find here a database name which you gave.



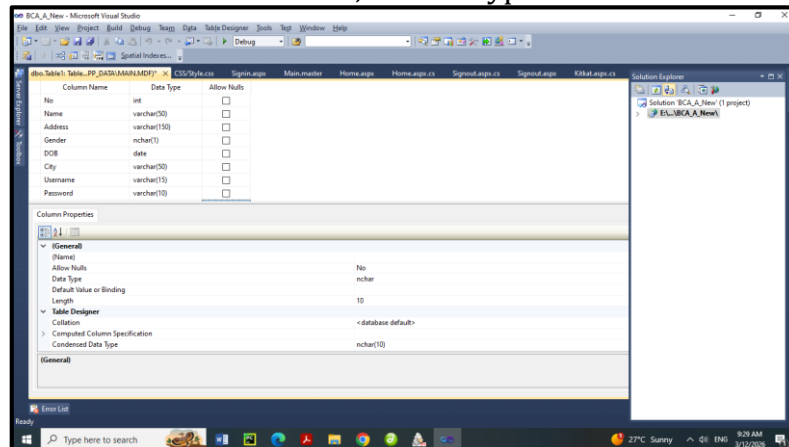
- Now in the above window you have to right click on the “Table” option to create a table. Then you will appear following menu.



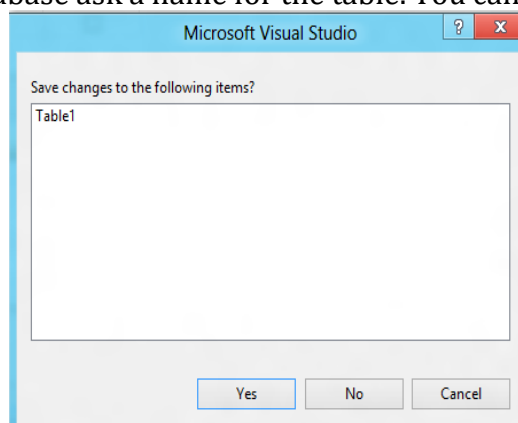
- Select “Add New Table” option from above menu.
- Now, you will appear the following screen.



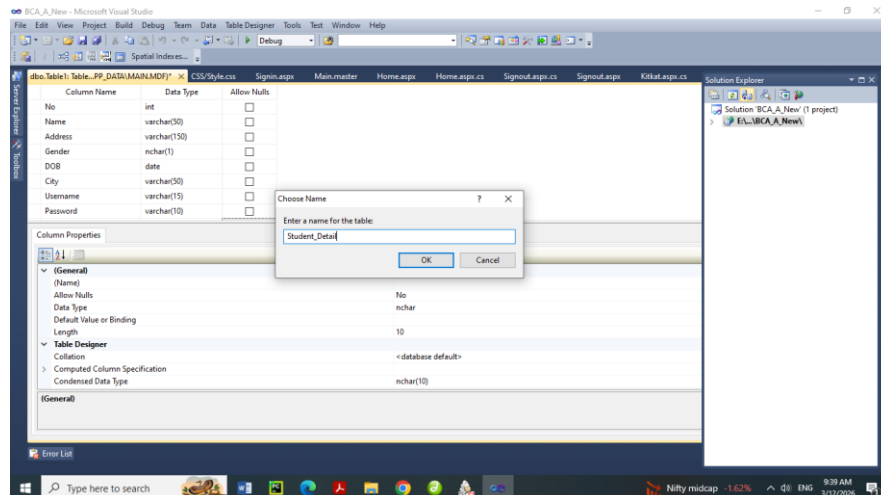
- Now, in the above screen define Field, its data type. Then close table tab.



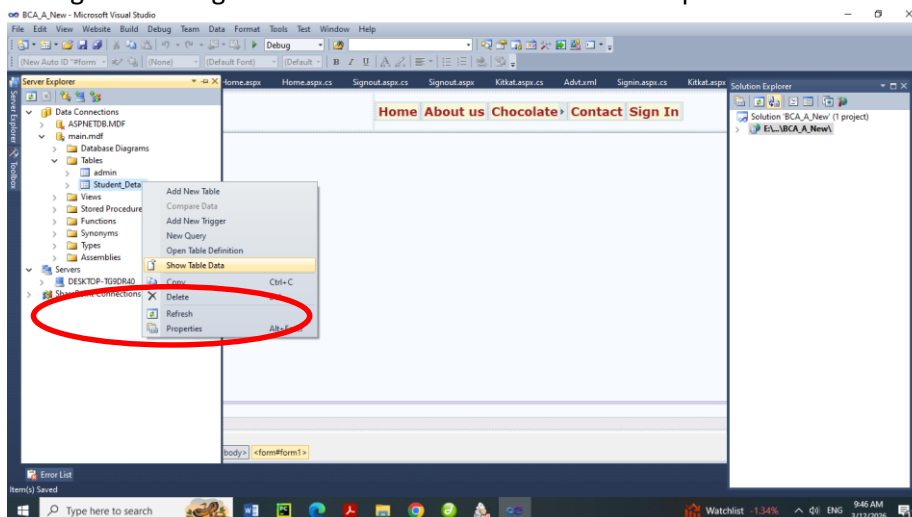
- So ,MSSQL server database ask a name for the table. You can see it in the following figure



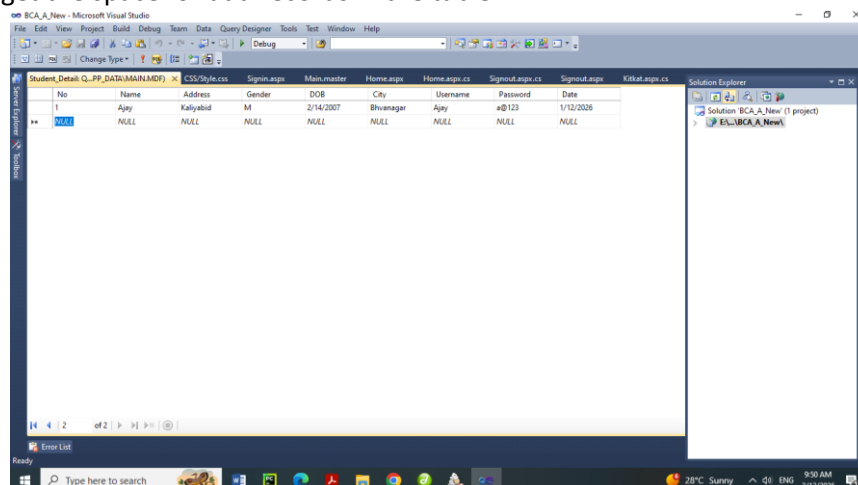
- Here you have to click on the “Yes” button. So, new dialog box will appear for new table name as below.



- Define new name and click on “Ok” button.
- Now, for adding records in the table. Right click on the table name just now create .you can see in the following table.
- Now, you will get following menu and select “Show Table Data” option.



- Now you will get the space for add records in the table.



- After entering data close the screen.

Working with data controls

Q.4 Explain Grid View Control in detail.

- ❑ The GridView control is a data bound control that display the values of a data source in the form of a table.
- ❑ In this table, each column represent a field and each row represent a record.
- ❑ The GridView control exists within the system.web.UI.WebControls namespace.
- ❑ The inheritance hierarchy of the GridView control is as follows :
 - System.object
 - System.web.UI.control
 - System.web.UI.webcontrols.webcontrol
 - System.web.UI.webcontrols.BaseDataBondControl
 - System.web.UI.webcontrols.DataBoundControl
 - System.web.UI.webcontrols.CompositeDataBoundControl
 - System.web.UI.webcontrols.GridView
- ❑ The GridView data control has a built in capability of sorting, paging, updating and deleting data.
- ❑ You can also set the column fields through the AutoGenerateColumns property to indicate whether bound fields are automatically created for each field in the data source.

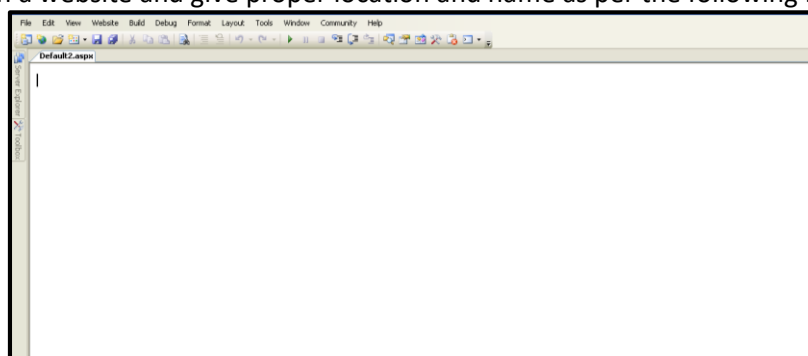
Column field types of the GridView class.

Field	Description
BoundField	Show the value of a field inn a data source.
Buttonfield	Shows a command button for each item in the GridView control.
CheckBoxField	Shows a CheckBox control for each item in the GridView control.
CommandField	Shows predefined command buttons to perform operations.
HyperLinkField	Shows the value of a field in a data source as a hyperlink.
ImageField	Sows an image of for each item in the GridView control.
TemplateField	Sows user defined content for each item inn the GridView control according to a specified template.

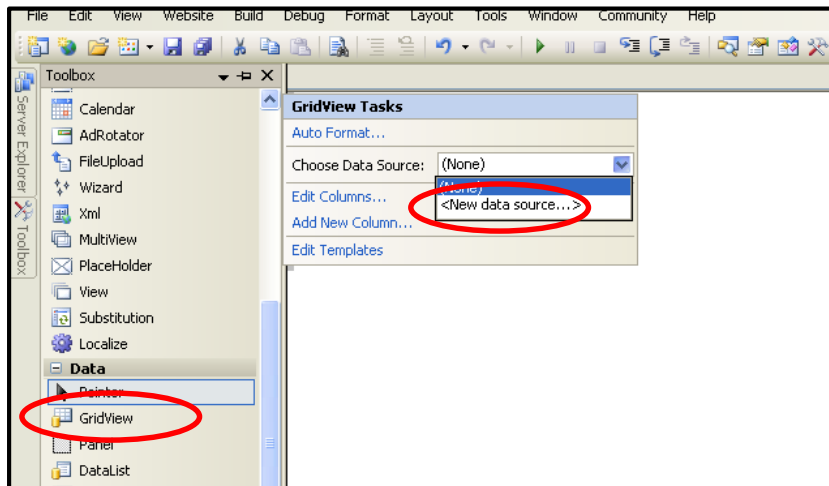
There are number of different properties to GridView. Among them some are below :

Property Name	Description
EditRowStyle	Performs the style settings for the row that is edited in the GridView control
EmptyDataRowStyle	Performs the style settings for the empty data row displayed in the GridView control.
FooterStyle	Performs the style setting for the footer row of the GridView control.
HeaderStyle	Performs the style setting for the header row of the GridView control.
AllowPaging	Set a value indicating whether the paging feature enabled.
AllowSorting	Set a value indicating whether the sorting feature enabled.

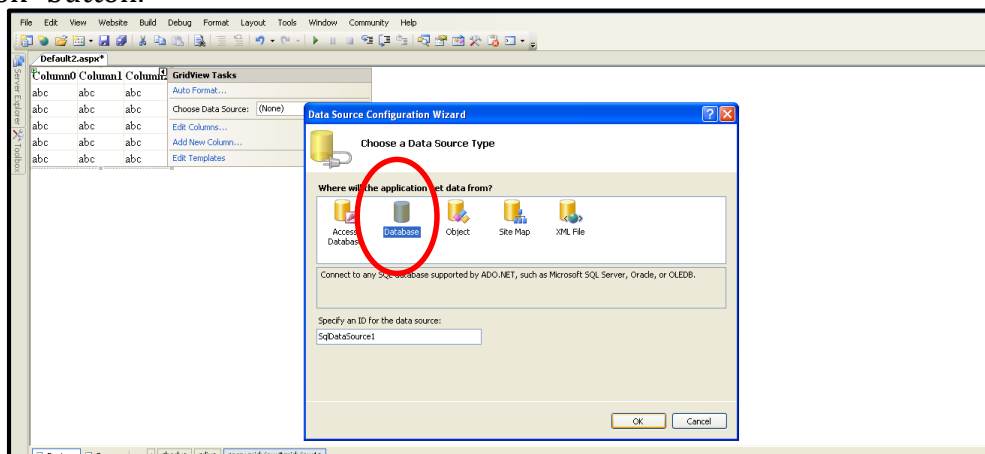
- ❑ First we open a website and give proper location and name as per the following figure.



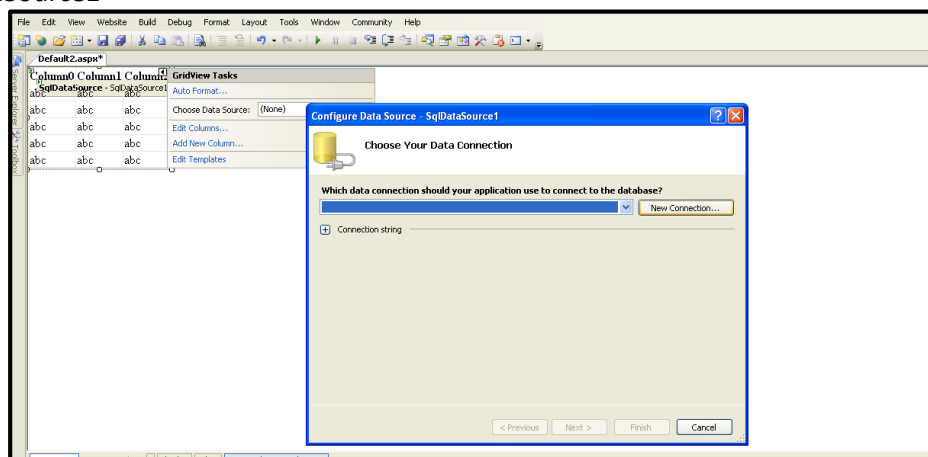
- Then you have to drag and drop the GridView data control on the web form in the design view, so it is look as below:



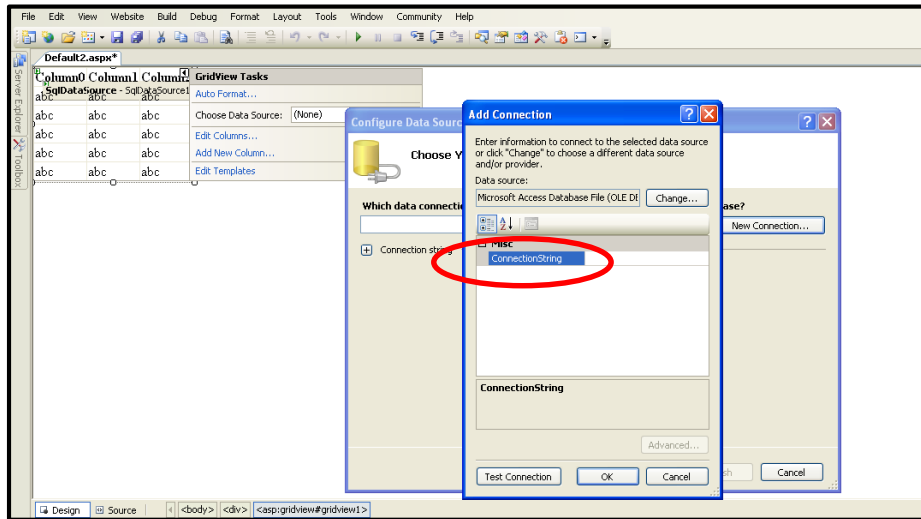
- Now select “New data source” from choose data source option which you can see the above figure, so you will get the following screen with the “Data Source Configuration Wizard” dialog box.
- From the “Data Source Configuration Wizard” you have to click on “Database” and click on “ok” button.



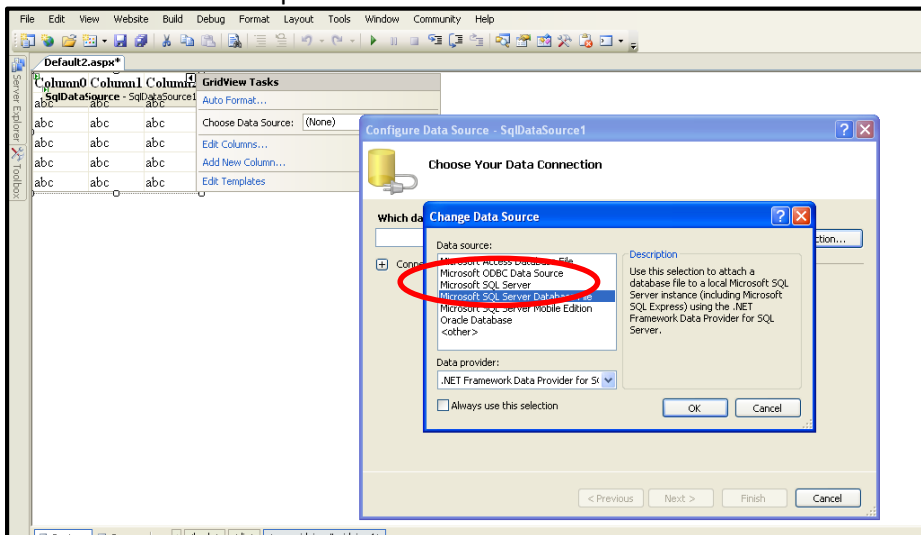
- So, you will appear following screen with new dialog box “Configuration Data Source – SqlDataSource1”



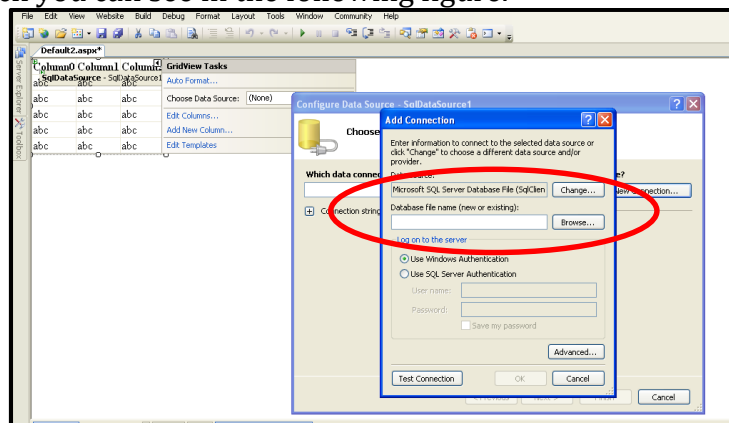
- Now, click on “New Connection” button from “Configuration Data Source – SqlDataSource1” dialogbox.



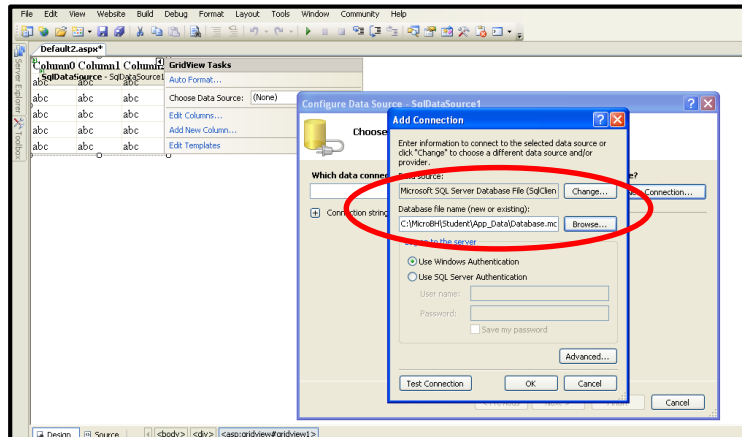
- So, you will appear new dialogbox “Add connection”. From this dialogbox, you have to click on “change” button to select data provider.



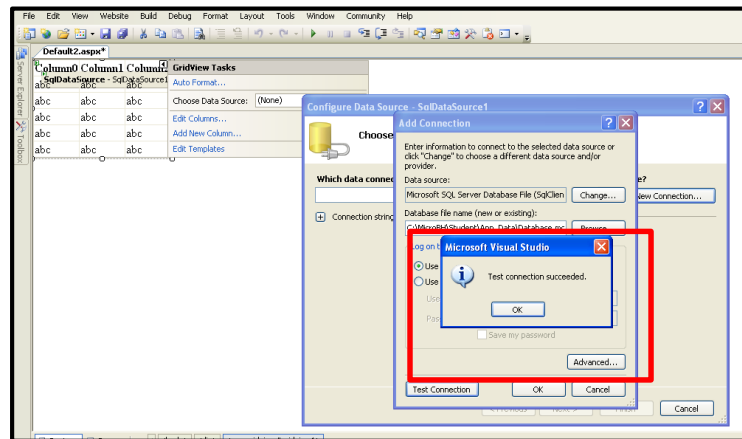
- This click open a new dialogbox “Change data source”, here you have to select “Microsoft SQL server database file” option and click on “ok” button.
- Again you will see the “Add connection” dialogbox with new text box and a button “Browse” which you can see in the following figure.



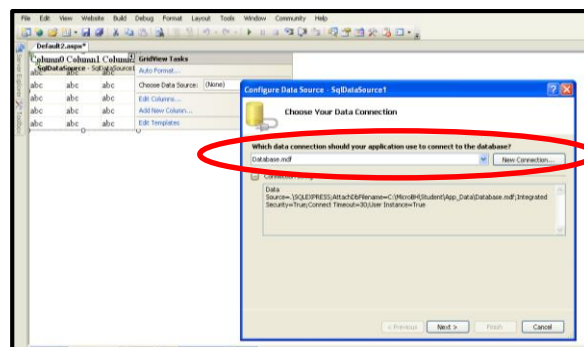
- Here you have to click on the “Browse” button and select appropriate database name, which contain the table of data.



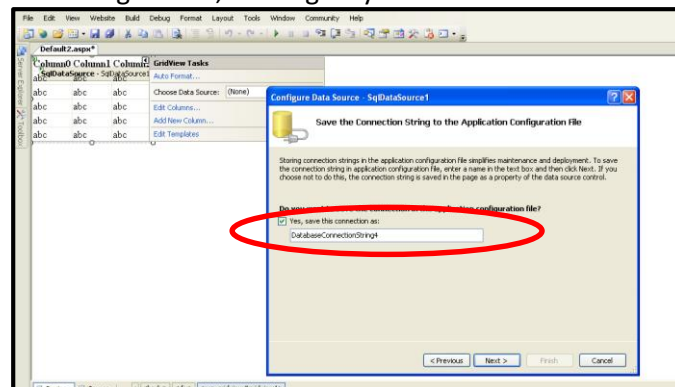
- After selecting database you have to click on the “Test Connection” button which you will appear at the button in the “Add Connection” dialogbox to check the connection. So you will get the following screen.



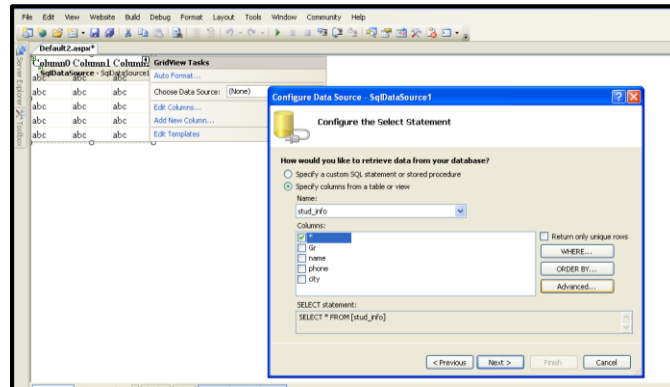
- Then click on the “Ok” button, you will appear following screen. Here you have to click on the “Next” button.



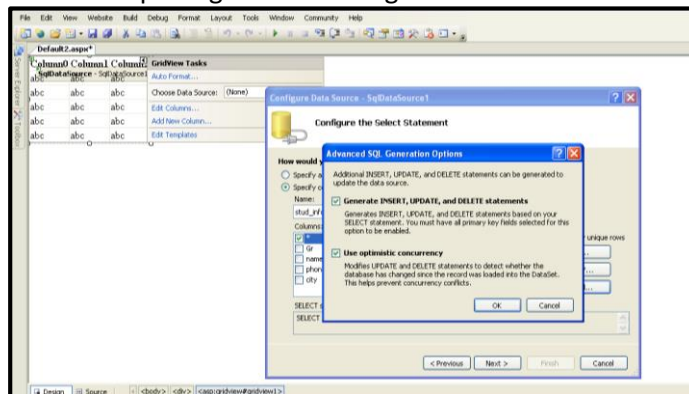
- So, you will get the following screen, here again you have to click on the “Next” button.



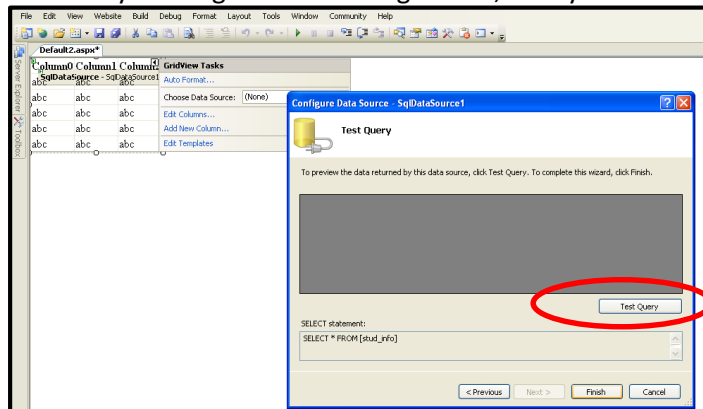
- Now you will get the following screen, here you have to select table name and different field which data you want to display in the GridView. Here if you select * option then all the field will select otherwise you have to check the individual field name. then click on the “Advanced...” button.



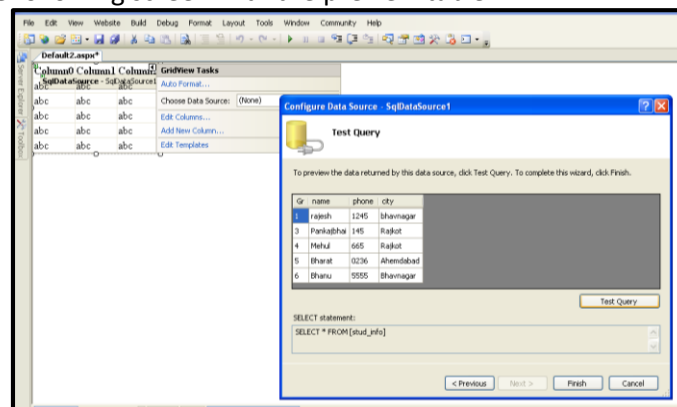
- So you will get the following screen with new dialogbox “Advanced SQL Generation Options”, here you have to select both the option given the dialogbox.



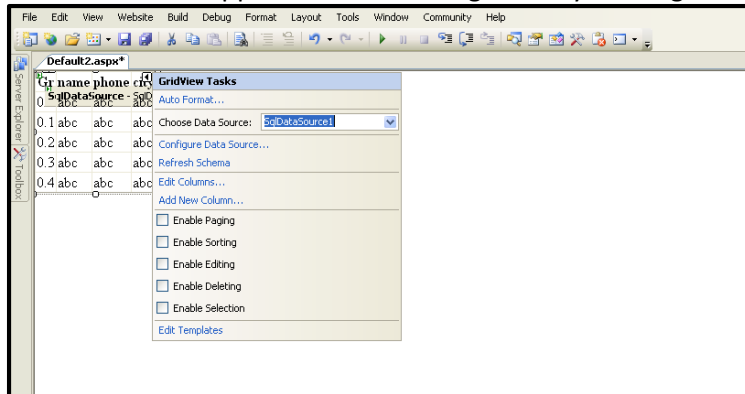
- Click on the “ok” button so you will get the following screen, here you have to click on the “Test Query”



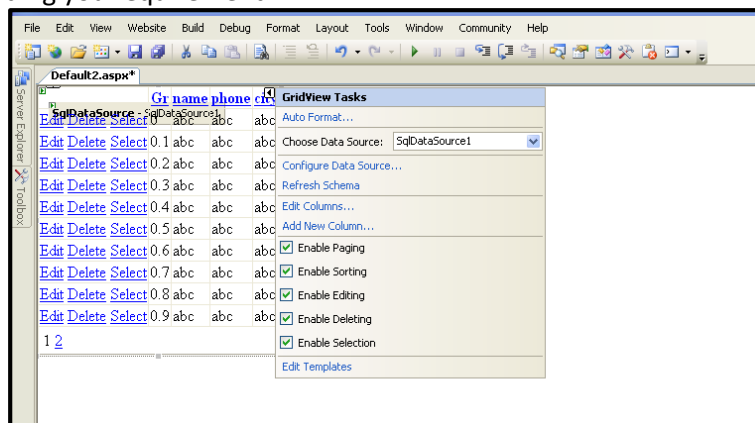
- So you will get the following screen with the preview table.



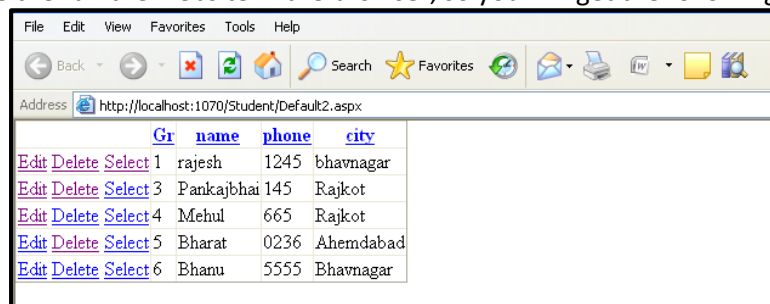
- Now click on the “finish” button appear in the above figure. So you will get the following figure



- Here you have to select Enable paging, Enable sorting, Enable Editing, Enable Deleting, Enable Selection according your requirement.



- Now, you have the run the website in the browser, so you will get the following screen.



Other Output

Action	Id	Name	LastName	Email
Edit Delete Select	1	Mark	Arton	markarton03@outlook.com
Edit Delete Select	2	Tony	Jacob	tonyjacob@yahoo.com
Edit Delete Select	3	Robert	Brown	robertbrown@yahoo.com
Edit Delete Select	4	Lucas	Biglia	lucasbiglia@gmail.com

List View Control:

- ❑ The ListView control is a data bound control used to display data from the associated data source.
- ❑ The ListView control enables a developer to display data in any format using templates and style.
- ❑ This control provides excellent customization and extensibility features.
- ❑ It provides the developer with control on the render html output and provides support for adding row, and sorting.
- ❑ The inheritance hierarchy of the ListView control is as follows:
 - System.object
 - System.Web.UI.Control
 - System.Web.UI.WebControls.WebControl
 - System.Web.UI.WebControls.BaseDataBoundControl
 - System.Web.UI.WebControls.DataBoundControl
 - System.Web.UI.WebControls.ListView

There are number of different properties of ListView. Among them some are follow:

Property Name	Description
AccessKey	Overrides the WebControl.AccessKey property. Setting this property is not supported by the ListView Control.
AlternatingItemTemplate	Sets the custom content for the alternating data item in a ListView control.
Controls	Obtains a ControlCollection object that represents the child controls of the ListView control.
CssClass	Override the WebControl.CssClass property. Setting this property is not supported by the ListView Control.
DataKeyNames	Sets an array that contains the names of the primary key fields for the items displayed in a ListView control.
DataKeys	Obtains a collection of Datakey objects that represent the datakey value for each item in a ListView control.

For Practical you have to following all the steps which is given in the GridView Control.

DataListControl:

- ❑ The DataList control is a data bound control that displays data by using templates. These templates define controls and html elements that should be displayed for an item.
- ❑ The DataList control exists within the system.Web.UI.WebControls names.
- ❑ The inheritance hierarchy of the DataList control is as follows:
 - System.object
 - System.Web.UI.Control
 - System.Web.UI.WebControls.WebControl
 - System.Web.UI.WebControls.BaseDataList
 - System.Web.UI.WebControls.DataList

Following templates are used in the DataList control:

Template Name	Description
AlternatingItemTemplate	Provides the format and content for alternating cell in the DataList control.
EditItemTemplate	Provides the format and content for the items to be edited in the DataList control.
FooterTemplate	Provides the format and content for the footer section of the DataList control, when defined, else, the footer section does not appear.

HeaderTemplate	Provides the format and content for the header section of the DataList control, else the header section does not appear.
ItemTemplate	Provides the format and content for items in the DataList control. It is a mandatory template.
SelectedItemTemplate	Provides the format and content for the presently selected item in the DataList control.
SeparatorTemplate	Provides the format and content for the separator separating items in the DataList control else the separator does not appear.

There number of properties of DataList Control. Among them some are following:

Property Name	Description
AlternatingItemStyle	Obtains the style properties for alternating items in the DataList control.
EditItemIndex	Sets the index number of the selected item in the DataList control to edit.
EditItemStyle	Set the style properties for the item selected for editing in the DataList control.
ExtractTemplateRows	Set a value that indicate whether rows of a table control, define in each template of a DataList control.
FooterStyle	Obtain the style properties for the footer section of the DataList control.
FooterTemplate	Sets the template for the footer section of the DataList control.
GridLine	Sets the grid line style to the DataList control.

For Practical you have to following all the steps which is given in the GridView Control.

Output

Student Details		
Student Id : 1 Name : Mark Last Name : Arton Email : markarton03@outlook.com	Student Id : 2 Name : Tony Last Name : Jacob Email : tonyjacob@yahoo.com	Student Id : 3 Name : Robert Last Name : Brown Email : robertbrown@yahoo.com
Student Id : 4 Name : Lucas Last Name : Biglia Email : lucasbiglia@gmail.com.com		